

WASP-12b

R-1764

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_160224 · created 2026-08-02T21:19:55+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457442.71331 +/- 0.00025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.091 +/- 0.031
Transit depth (Rp/Rs)^2	0.82 +/- 0.56 [%]
Orbital Inclination (inc)	86.13 +/- 2.71
Airmass coefficient 1 (a1)	1.027 +/- 0.015
Airmass coefficient 2 (a2)	-0.0193 +/- 0.0069
Scatter in the residuals of the lightcurve fit is	1.44 %
Best Comparison Star	#1 - [303, 261]
Optimal Aperture	3.67
Optimal Annulus	9.84
Transit Duration (day)	0.1109 +/- 0.0025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.091 $\pm$ 0.031 vs 0.1178 (deviation 0.82 σ)
Duration fit vs published	0.1109 d vs 0.125 d (deviation 5.37 σ)
Mid-time O-C	-0.7 min
Depth SNR	2.8
Residual scatter	1.44 %

Light curve

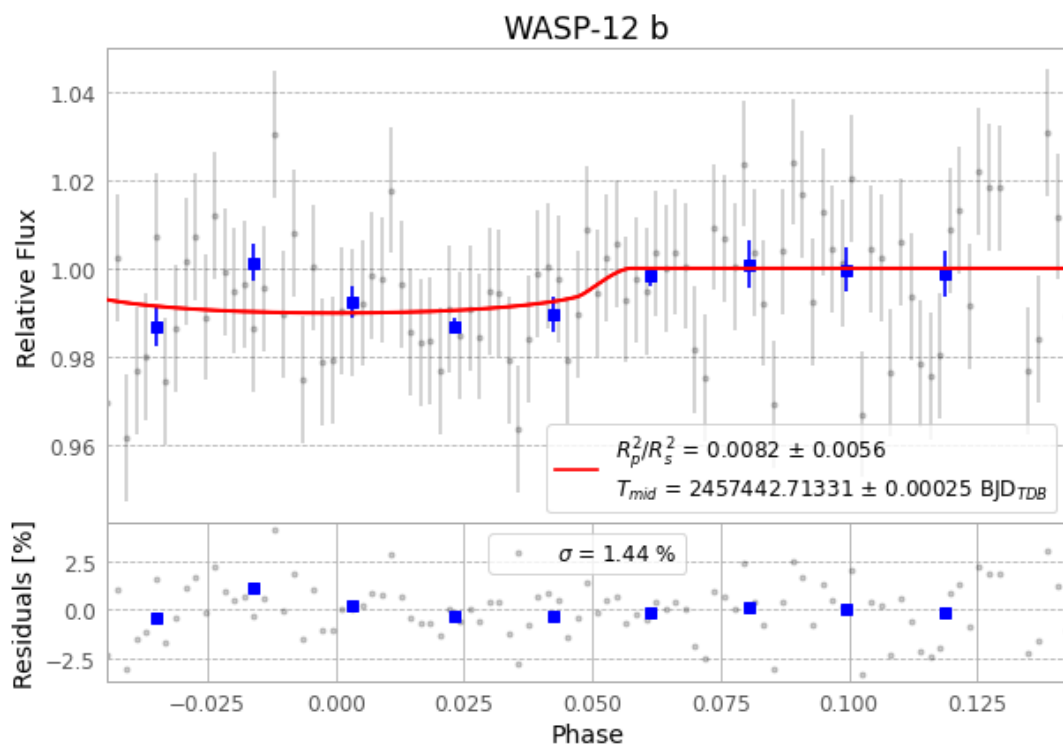


Figure 1 — FinalLightCurve_WASP-12 b_2016-02-23.png (SHA-256 1ce7118de4af0c8f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [320, 230], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0a4a44e9ebfa167e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

99 FITS frames (65 MB), WASP-12160224035112.FITS → WASP-12160224084512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-160224.log (37240f0da1e6...), FinalLightCurve_WASP-12 b_2016-02-23.png (1ce7118de4af...), FinalLightCurve_WASP-12 b_2016-02-23.csv (c18afb0b9435...)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 21:19Z.